Daniel E. Martínez

Professor of Biology and Molecular Biology

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Faculty Positions

2011-	**Professor,** Pomona College
2009-11	**Biology Department Chair,** Pomona College
2002-11	**Associate Professor,** Pomona College
2004-06	**Coordinator Molecular Biology Program**, Pomona College
1997-02	**Assistant Professor,** Pomona College
1996-97	**Assistant Professor,** St. Lawrence University, Canton NY

Postdoctoral Positions

1995-96	**Postdoctoral Fellow**, Developmental Biology Center, University of California, Irvine. Lab: Dr. Hans R. Bode
1993-95	**Alfred P. Sloan Postdoctoral Fellow**, Developmental Biology Center, University of California, Irvine.

Education

1993	**Ph.D.** in Ecology and Evolution, State University of New York, Stony Brook
1982	**Licenciado**, Universidad Nacional del Sur, Bahía Blanca, Argentina

Most Recent Grants and Awards

2010-17	NIH, RO1 (Mechanisms of Aging in Hydra, $1,263,358)
2013-15	UC Riverside/Templeton Foundation, The Immortality Project ($250,091)
2015	HHMI Collaborative Grant ($13,000; with Dr. Cathy McFadden, HMC)
2013-14	Max Planck Institute For Demographic Research ($80,000)
2009-10	Max Planck Institute For Demographic Research ($36,000)
2008-09	Max Planck Institute For Demographic Research ($33,000)
2007-08	Max Planck Institute For Demographic Research ($33,000)
2006-07	Max Planck Institute For Demographic Research ($31,270)
2005-06	Max Planck Institute For Demographic Research ($31,270)
2005-09	NSF, Assembling the Tree of Life (Total $2,994,192; Pomona: $69,590)
2004-05	Max Planck Institute For Demographic Research ($22,275)
2003-05	NSF, Mechanism of Development Program ($152,177)
2003	Mellon Foundation Semester Research Leave grant ($35,500)
2003	Mellon Faculty Research Partnership Program ($3,900)
2002-03	Hirsch Research Grant ($9,400)
2002	NSF REU Supplement ($12,000)
2002	NSF ROA ($19,000)
2001-04	NSF (coll. Dr. H. Bode, UC Irvine, $1,300,000)
1998-02	US-Israel Binational Sci. Foundation (coll. Dr. O. Mokady, Tel-Aviv U. $150,000)

Selected Publications (* undergraduate coauthor)

Campos, S.E., S. Naziri, J. Crane, J. Tsverov, B. D. Cox, C. Ciampa, R. B. Spencer*, R.E. Steele, D. E. Martínez & C. E. Juliano. 2025. Wnt signaling restores evolutionary loss of robust foot regeneration rates in Hydra. *Nature Communications* **16**, Article number: 11447

Kon-Nanjo, K., Kon, T., Koubkova Y., Tracy C-Ting, Rodriguez-Terrones, D., Falcon, F., Martínez, D. E., Steele, R. E., Tanaka, E. M., Holstein, T. W., Simakov, O. 2025. The dynamic genomes of Hydra and the anciently active repeat complement of animal chromosomes. Genome Biol. **26**:186.

Makamure, C. E., S. Justinen, D. E. Martínez, D. A. Hessinger, I. Kremsky, 2025. Cool temperature inhibits binary fission and results in phenotypic and transcriptomic changes that suggest inducible aging in *Diadumene lineata*. *BMC Research Notes,* **18**:293

Lu A., M. Kimble, S. Justinen, D. P. Morris, C. Wang, D. E. Martínez, D. A. Hessinger, 2023. BK Channels Function in Nematocyst Discharge from Vibration-Sensitive Cnidocyte Supporting Cell Complexes of the Sea Anemone *Diadumene lineata. Biological Bulletin,* **245***.*

Moreau X., M. Claeys-Bruno, J-P. Andraud, H. Macarie, D. E. Martínez, M. Robin, M. Sergent, L. De Jong, 2022. Hydra bioassay for the evaluation of chlordecone toxicity at environmental concentrations, alone or in complex mixtures with dechlorinated byproducts: experimental observations and modeling by experimental design. *Environmental Science and Pollution Research* **29**: 91017–91035. https://doi.org/10.1007/s11356-022-22050-8

Simakov O., J. Bredeson, K. Berkoff, F. Marletaz, T. Mitros, D. T. Schultz, B. L. O'Connell, P. Dear, D.E. Martinez, R. E. Steele, R. E. Green, C. N. David, D. S. Rokhsar, 2022. Deeply conserved synteny and the evolution of metazoan chromosomes. *Science Advances* **8** (5): 1-17

Wong, W.Y., O. Simakov, D. M. Bridge , P. Cartwright , A. J. Bellantuono , A. Kuhn , T. Holstein , C. David , R. E. Steele , D. E. Martínez, 2019. Expansion of a single transposable element family is associated with genome size increase and radiation in the genus Hydra. *Proc. Natl. Acad. Sci. USA.* **116** (46): 22915–22917.

Steele, R.E., Updegrove, M.D., Kirolos, S.A., Mowery, L., Martínez, D.E., Bryant, P.J. 2019. Reproductive bet-hedging and existence in vernal pools as components of Hydra life history. *Biol. Bull*. **237**: 111–118.

Schaible, R., A. Scheuerlein, M. J. Danko, J. Gampe, D. E. Martínez, & J. W. Vaupel, 2015. Constant mortality and fertility over age in Hydra. *Proc. Natl. Acad. Sci. USA.* **112** (51): 15701–15706.

Bellantuono, A. J., D. M. Bridge & D. E. Martínez, 2015. *Hydra* as a tractable, long-lived model system for senescence. *Invertebrate Reproduction & Development* **59** (S1): 39-44.

Campbell R. D., A. R. Iñiguez*, A. J. Iñiguez* & D. E. Martínez, 2013. Hydra of Hawaii: Phylogenetic Relationships with Continental Species. *Hydrobiologia* **713**: 199-205.

Levitis, D. A & D. E. Martínez, 2013. The two halves of U-shaped mortality. *Frontiers in Genetics of Aging,* **4**: 1-6.

Rose M. R., T. Flatt, J. L. Graves, L. F. Greer, D. E. Martínez, M. M. Matos, L. D. Mueller, R. J. Shmookler Reis & Parvin Shahrestani, 2012. What is Aging? *Frontiers in Genetics of Aging* **3**: 1-3.

Martínez, D E & D. Bridge, 2012. *Hydra*, the everlasting embryo, confronts aging. *Int. J. Dev. Biology*. **56**: 411-423

Juárez A.B., C.G.Vélez. A. R. Iñiguez*, D. E. Martínez, M. C. Rodríguez, M. S. Vigna & M. C. Ríos de Molina, 2011. A *Parachlorella kessleri* (Trebouxiophyceae, Chlorophyta) strain from an extremely acidic geothermal pond in Argentina. *Phycologia* **50**: 413-421.

Bridge D., A. Theofiles*, R. Holler*, E. Marcinkevicius*, R. Steele, & D. Martínez, 2010. FoxO and Stress Responses in the Cnidarian *Hydra vulgaris*. *PLoS ONE* **5** (7): e11686

Martínez D E , A R Iñiguez*; K M Percell*; J B Willner*; J Signorovitch*; R D Campbell, 2010. Phylogeny And Biogeography Of Hydra (Cnidaria: Hydridae) Using Mitochondrial And Nuclear DNA Sequences. *Molecular Phylogenetics and Evolution* **57** (20): 403-410.

Chapman, J.A., Kirkness, E.F., Simakov, O., Hampson, S.E., Mitros, T., Weinmaier, T., Rattei, T., Balasubramanian, P.G., Borman, J., Busam, D., Disbennett, K., Sumin, N., Sutton, G.G., Viswanathan, L.D., Walenz, B., Goodstein, D.M., Hellsten, U., Kawashima, T., Prochnik, S.E., Putnam, N.H., Shu, S., Blumberg, B., Dana, C.E., Gee, L., Kibler, D.F., Law, L., Lindgens, D., Martinez, D.E., Peng, J., Wigge, P.A., Bertulat, B., Guder, C., Nakamura, Y., Ozbek, S., Watanabe, H., Konstantin, Khalturin, Hemmrich, G., Franke, A., Augustin, R., Fraune, S., Hayakawa, E., Hayakawa, S., Hirose, M., Hwang, J.S., Kazuholkeo, Nishimiya-Fujisawa, C., Ogura, A., Takahashi, T., Steinmetz, P.R.H., Zhang, X., Aufschnaiter, R.,

Eder, M.-K., Gorny, A.-K., Salvenmoser, W., Heimberg, A.M., Wheeler, B.M., Peterson, K.J., Böttger, A., Tischler, P., Wolf, A., Gojobori, T., Remington, K.A., Strausberg, R.L., Venter, J.C., Technau, U., Hobmayer, B., Bosch, T.C.G., Holstein, T.W., Fujisawa, T., Bode, H.R., David, C.N., Rokhsar, D.S., Steele, R.E., 2010. The dynamic genome of *Hydra*. *Nature* **464**: 592-596.

Acuña, F.H., A. C. Excoffon, S. R. McKinstry* & D. E. Martínez, 2007. Characterization of *Aulactinia* (Actiniaria: Actiniidae) species from Mar del Plata (Argentina) using morphological and molecular data. *Hydrobiologia*, **592**: 249-256.

Bridge D. M., C. T. Ha, A. Nemir*, A. Renden*, M. M. Rorick*, A. Shaffer*, D. M. Underkoffler*, A. E. Wills*, & D. E. Martínez, 2004.Variations on a theme? Polyp and medusa development in *Podocoryna carnea. Hydrobiologia,* **530-531**: 299-307

Elías R., O. Iribarne, Bremec C. S. & D. E. Martínez, 2004. Comunidades bentónicas de fondosblandos. In Piccolo M. C., Hoffmeyer M. S. (Eds.) *Ecosistema del estuario de Bahía Blanca*. Instituto Argentino de Oceanografía, Argentina.

Bremec C. S., D. E. Martínez & R. Elías, 2004. Asociaciones bentónicas de fondos duros y comunidades incrustantes. In Piccolo M. C , Hoffmeyer M. S. (Eds.) *Ecosistema del estuario de Bahía Blanca*. Instituto Argentino de Oceanografía, Argentina.

Martínez, D. E., 2002. Zootypes. In Mark Pagel (Ed*.) Encyclopedia of Evolution.* Oxford University Press (Oxford).

Martínez, D. E., 2002. Senescence and Rejuvenation in Asexual Metazoans. In R. N. Hughes (Ed*.) Reproductive Biology of Invertebrates, Vol. XI, Progress in Asexual Reproduction.* Oxford & IBH Publishing (New Delhi). Pp: 115- 140.

Kaestner K. H., W. Knöchel, & D. E. Martínez, 2000. Unified nomenclature for the Winged Helix/Forkhead transcription factors. *Genes & Development* **14**: 142-146.

Levinton J., P. Klerks, D.E. Martínez, C. Montero, C. Sturmbauer, L. Suatoni, W. Wallace, 1999. Running the gauntlet: pollution, evolution and reclamation of an estuarine bay. In Whitfield, M., Matthews, J. Reynolds, C. (Eds*.) Aquatic Life Cycle Strategies: Survival in a Variable Environment.* Marine Biological Association of the United Kingdom, Plymouth.

Bode H., D. Martínez, M. A. Shenk, K. Smith, R. Steele & U. Technau, 1999. Evolution of Head Development**.** *Biological Bulletin*, **196**:408-410.

Martínez D. E., D. Bridge, L. Masuda-Nakagawa & P. Cartwright, 1998. Cnidarian homeoboxes and the zootype. *Nature* **393**:748-749.

Martínez D. E., 1998. Mortality patterns suggest lack of senescence in hydra. *Experimental Gerontology* **33:** 217-225.

Martínez D.E., M-L. Dirksen, P. M. Bode, M. Jamrich, R. E. Steele & H. R. Bode, 1997. *Budhead*, a fork head/HNF-3 homolog, is expressed during axis formation and head specification in hydra. *Developmental Biology* **192**: 523-536.

Martínez D. E., 1996. Rejuvenation of the disposable soma: Repeated injury extends lifespan in an asexual annelid. *Experimental Gerontology* **31**: 699-704.

Martínez D. E. & J. S. Levinton, 1996. Adaptation to heavy metals in the aquatic oligochaete *Limnodrilus hoffmeisteri*: evidence for control by one gene. *Evolution* **50**: 1339-1343.

Levinton J. S., D. E. Martínez, M. M. McCartney & M. L. Judge, 1995. The effect of water flow on movement, burrowing, and distributions of the gastropod *Ilyanassa obsoleta* in a tidal creek. *Marine Biology* **122**: 417-424.

Cheng I-J, J. S. Levinton, M. McCartney, D. Martínez & M. J. Weissburg, 1993. A bioassay approach to seasonal variation in the nutritional value of sediment. *Marine Ecology Prog. Ser.* **94**: 275-285.

Martínez D. E. & J. S. Levinton, 1992. Asexual metazoans undergo senescence. *Proc. Natl. Acad. Sci. USA.* **89**: 9920-9923.

Bastida R., A. Roux & D. E. Martínez, 1992 Benthic communities of the Argentine Continental Shelf. *Oceanologica Acta* **15** (6): 687-698.

Invited Speaker

Aquatic Research Models to Study Regeneration and Aging, Nice, France, Dec 16-17, 2019 [**Keynote Speaker**]
Hydra module of Neural Systems and Behavior Course, Marine Biological Laboratory, Woods Hole, July 2019.
Campus Agro-Environnemental Caraïbe, Martinique, May 27, 2019.

Loma Linda University, October 4, 2018.
Symposium on "Implications Of Mechanisms In Invertebrate Organisms For Mammalian Health Span And Longevity", Naples-Florence, Italy, Aug 28 - Sep 2, 2017.
First Biennial North American Meeting of Hydroid Biologists, Bodega Bay CA, Sep 22-25, 2016.
Atheists United, Los Angeles, June 28, 2015.
Immortality Project Capstone Conference, UC Riverside, Riverside CA, May 2015.
Second Annual Meeting of the Evolutionary Demography Society, Stanford U., Palo Alto CA, Nov 10-12, 2014.
Immortality Project Midpoint Conference, UC Riverside, Riverside CA, June 2014.
First Annual Meeting of the Evolutionary Demography Society, Odense, Denmark, October 6-10, 2013.
Symposium on Model Invertebrate Systems for the Study of Stem Cells in Development, Detroit, MI, July 14 - 19, 2013.
Pauley Summer Program in Marine Biology, Hawaiian Institute of Marine Biology, June 5, 2013.
Eugene and Ruth Roberts Summer Student Academy at City of Hope, Duarte CA, June 27, 2013.
Rancho Santa Anna Botanical Gardens, Claremont CA. October 21, 2011.
International Workshop, Evangelische Akademie Tutzing, Germany, Sep. 12-15, 2011.
Fifth Annual Division of Aging Biology New Investigators Forum. NIH, Bethesda MD, June 27, 2011.
Max Planck Institute for Demographic Research (MaxNetAging program), March 1, 2011.
Grantees Meeting: Invertebrates Model of Aging. NIH, Bethesda MD, Nov. 5, 2010.
Universidad Nacional de Buenos Aires, Buenos Aires, Argentina, October 4, 2010.
International Workshop, Evangelische Akademie Tutzing, Germany, Sep. 14-17, 2009.
California Polytechnic Institute, Pomona, CA, July 30, 2009.
University of Nebraska, Lincoln. April 29, 2009.
California Polytechnic Institute, Pomona, CA, Nov. 2008.
Fukuoka Women's University, Fukuoka, Japan, 2008.
Kyushu University, Fukuoka, Japan, 2008.
National Institute of Genetics, Mishima, Japan, 2008.
3rd Annual Cnidarian Tree of Life Meeting (La Paz, Mexico, July 19-24, 2008).
Cnidarian Tree of Life meeting (St. John, US Virgin Islands, July5-10, 2007) .
2nd International Research Workshop of Evol. Demography; Max Planck I. D. R., Rostock, Germany, 2006.
Evolutionary Demography Meeting, University of Virginia, Charlottesville VA, 2005.
Workshop on Vital Age-Trajectories: Realities and Theories; Max Planck I. D. R., Rostock, Germany, 2004.
Max Planck Institute for Demographic Research, Rostock, Germany, 2004.
Laboratorio Ecotono, CRUB, Universidad Nacional del Comahue, Argentina, 2004.
7th International Conference on Coelenterate Biology, Lawrence, Kansas; 2003.
Workshop on Cnidarian Development and Evolution, University of California at Irvine, 2002.
Dipartimento di Biologia, Universitá degli Studi di Lecce, Italy, 2000.
Cal State Fullerton, Fullerton CA, 2000.
Beckman Research Institute, City of Hope, Duarte CA, 1999.
Institute for Nature Conservation Research, Tel Aviv University, Tel Aviv, Israel, 1999.
8th International Workshop on Hydroid Development, Tutzing, Germany, 1999.
School of Biological Sciences, University of Nebraska, Lincoln, 1999.
International Meeting on Forkhead/Winged Helix Proteins, La Jolla CA, 1998.
Gordon Conference, Biology of Aging, Il Ciocco, Italy, 1998 (**keynote speaker**).
7th International Workshop on Hydroid Development, Tutzing, Germany, 1997.
Pomona College, Claremont CA, 1997.
West Coast Regional Developmental Biology Conference, Fallen Leaf Lake CA, 1996.
St. Lawrence University, Canton, NY, 1996.
6th International Workshop on Hydroid Development, Tutzing, Germany, 1995.
Scripps Institution of Oceanography, La Jolla CA, 1995.
California State University, San Marcos CA, 1995.
Department of Ecology and Evolutionary Biology, UCI, Irvine CA, 1994.
5th International Workshop on Hydroid Development, Günzburg, Germany, 1993.
Dipartimento di Biologia, Universitá degli Studi di Padova, Italy, 1993.

Journal Reviews Requested

Bioessays	Clinical Genetics
Functional Ecology	Blackwell
Development, Genes and Evolution	Biological Bulletin
Journal of Molecular Evolution	Genetics
Quarterly Review of Biology	Biological Reviews
Hydrobiologia	PLoS ONE
Evolution and Development	Develpmental Biology

Professional Service

2013	Reviewer Integrative Organismal Systems of the NSF.
2012-	Editorial Board of Frontiers in Genetic of Aging (Review Editor).
2012	Reviewer for the Division of Integrative Organismal Systems of the NSF.
2010	Panel reviewer for the Ford Foundation Predoctoral Fellowship for Minorities Program. National Research Council, Washington DC. March.
2010	Reviewer for Biodiversity Surveys and Inventories panel of the National Science Foundation.
2009	Panel reviewer for the Ford Foundation Predoctoral Fellowship for Minorities Program. National Research Council, Washington DC. March.
2008	Panel reviewer for the Ford Foundation Predoctoral Fellowship for Minorities Program. National Research Council, Washington DC. March.
2007	Panel reviewer for the Ford Foundation Predoctoral Fellowship for Minorities Program. National Research Council, Washington DC. March.
2006	Reviewer for the Developmental Mechanism Program of the National Science Foundation.
2005	Reviewer for the Developmental Mechanism Program of the National Science Foundation.
2004	Reviewer for the Developmental Mechanism Program of the National Science Foundation.
2003	Reviewer for US-Israel Binational Science Foundation
2002	Panel reviewer for the Ford Foundation Predoctoral Fellowship for Minorities Program. National Research Council, Washington DC. March 8-9.
2001	Panel reviewer for the Ford Foundation Predoctoral Fellowship for Minorities Program. National Research Council, Washington DC. March 9-10.
2000	Reviewer for the Developmental Mechanism Program of the National Science Foundation.
2001	Featured on a three-program series on aging entitled: "Living Better, Living Longer" Aired on May 2nd in the Discovery Health Channel.
2000-	Specialist advisor for the HUGO Gene Nomenclature Committee (http://www.genenames.org/cgi-bin/genefamilies/set/508)
1999-	Fox Nomenclature Editor: personally responsible for assigning names to newly discovered Winged-Helix Proteins. (http://www.biology.pomona.edu/fox.html)
1995	Creator of and responsible for the WWW site “The Hydra Library” (http://pages.pomona.edu/~dmartinez/library.html)
1984-86	Editor of *BIOMA* (ISSN 0326-6303), a conservationist magazine.

Articles and Videos Discussing the Hydra Aging Research

1. Nathaniel Richnov. Can a Jellyfish Unlock the Secret of Immortality? *New York Times Magazine*. 28 Nov 2012

   http://www.nytimes.com/2012/12/02/magazine/can-a-jellyfish-unlock-the-secret-of-immortality.html

2. Lise Brix. Her er dyret, som kan leve for evigt.*Videnskab.dk*. 17 Oct 2013 (in Danish)

http://videnskab.dk/miljo-naturvidenskab/her-er-dyret-som-kan-leve-evigt

3. Robert Krulwich. The Animal That Wouldn't Die. *National Public Radio (NPR).* 26 Sept 2014 (video)

   https://www.youtube.com/watch?v=GRNVc2xC2O0

4. Stephanie Pappas. Hail the Hydra, an Animal That May Be Immortal. *Live Science*. 22 Dec 2015

   https://www.livescience.com/53178-hydra-may-live-forever.html

5. Susan Milius. Organisms age in myriad ways — and some might not even bother. ScienceNews. 13 Jul 2016

   https://www.sciencenews.org/article/organisms-age-myriad-ways-and-some-might-not-even-bother

6. Jon Cohen. Death-Defying Experiments. *Science* **350** (6265): 1186-1187. 4 Dec 2015

   http://science.sciencemag.org/content/350/6265/1186

7. Lise Brix. Nu er det bevist: Dette dyr ældes ikke.*Videnskab.dk*. 7 Dec 2015 (in Danish)

   http://videnskab.dk/miljo-naturvidenskab/nu-er-det-bevist-dette-dyr-aeldes-ikke

8. Oscar Cusó. Por qué la hidra es 'inmortal'. *El País*. 31 Oct 2016 (in Spanish)

   http://elpais.com/elpais/2016/10/06/ciencia/1475752826_096528.html

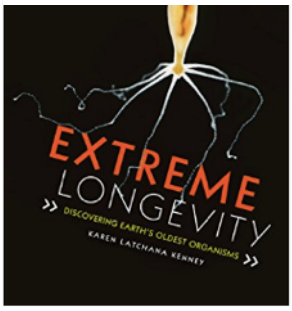
9. Maria A. Blasco & Mónica G. Salomone, 2016. Morir joven, a los 140. Editorial Paidos. (in Spanish)



10. Red Bull TV series Liquid Science. Season 1, episode 2: Living Forever. From minuet ~10

    https://www.redbull.tv/en_US/page/rrn:content:episode-videos:6700191b-7b12-5652-acae-9d5c54e38be5?cid=top_results

11. Book by Karen Kenney entitled "Extreme Longevity: Discovering Earth's Oldest Organisms". Twenty-First Century Books (2018).



12. Jessica Mouzo. En busca del elixir de la juventud: así avanza la carrera científica contra el envejecimiento. El Pais 12 Nov 2022 (in Spanish)

    https://elpais.com/salud-y-bienestar/2022-11-13/en-busca-del-elixir-de-la-juventud-asi-avanza-la-carrera-cientifica-contra-el-envejecimiento.html)